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Enhanced Detection of Sleep Disorders Using Hybrid Deep Learning and Ensemble Boosting: CNN-LSTM, XGBoost, CatBoost, AdaBoost, and LightGBM

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Abstract

Sleep disorders — Insomnia and Sleep Apnea — affect over one billion people globally and severely impact quality of life. Prior work achieved a best accuracy of 90.66% using Random Forest on the Sleep Health and Lifestyle Dataset. This study presents a comprehensive benchmark of eleven classifiers: five classical ML models, four gradient-boosting ensembles (XGBoost, CatBoost, AdaBoost, LightGBM), an LSTM, and a proposed CNN-LSTM hybrid. Using 5-fold stratified cross-validation, SMOTE oversampling, and SHAP explainability, the CNN-LSTM achieved 96.8% accuracy, followed by CatBoost (95.1%) and LightGBM (94.7%). All boosting ensembles outperformed the Random Forest baseline. SHAP analysis confirmed sleep duration, sleep quality, and BMI as dominant predictors.

Keywords: Sleep Disorders, Insomnia, Sleep Apnea, CNN-LSTM, XGBoost, CatBoost, AdaBoost, LightGBM, SHAP, SMOTE.

I. INTRODUCTION

Sleep disorders, particularly Insomnia and Obstructive Sleep Apnea (OSA), represent a critical public health burden affecting an estimated 50–70 million Americans and over one billion individuals globally. Chronic sleep deprivation is linked to cardiovascular disease, type-2 diabetes, obesity, anxiety, and cognitive decline. Conventional diagnosis relies on expensive polysomnography (PSG). Machine learning offers a scalable, cost-effective alternative using lifestyle and physiological data.

Payne et al. benchmarked five classical classifiers on the Sleep Health and Lifestyle Dataset, achieving a best accuracy of 90.66% with Random Forest. However, no study on this dataset has explored gradient-boosting ensembles or deep learning architectures. This work fills that gap by systematically evaluating eleven classifiers including XGBoost, CatBoost, AdaBoost, LightGBM, LSTM, and a CNN-LSTM hybrid, with 5-fold cross-validation, SMOTE class balancing, and SHAP interpretability.

Fig. 8: End-to-End Classification Workflow



Fig. 1: End-to-End Classification Workflow

II. RELATED WORK

Sham Fang Ying et al. connected sleep disorders to lifestyle via SVM at 72.73%. Sanchez-Trigo et al. applied Random Forest at 74%. Alan Schwartz et al. used ElasticNet achieving 73.2% for sleep apnea and 72.9% for insomnia. Ricardo Buettner et al. demonstrated EEG-based RF classification at 90.12%. Payne et al. benchmarked five classical classifiers, with Random Forest achieving 90.66%.

Gradient-boosting methods consistently outperform classical RF on structured biomedical data. Chen and Guestrin introduced XGBoost with L1/L2 regularization. Prokhorenkova et al. proposed CatBoost with ordered boosting for categorical features. Ke et al. introduced LightGBM for histogram-based efficient learning. Freund and Schapire introduced AdaBoost for iterative reweighting of misclassified samples. No prior work applies these to the Sleep Health and Lifestyle Dataset.

III. DATASET AND PREPROCESSING

A. Dataset Description

The Sleep Health and Lifestyle Dataset (Tharmalingam, 2023) from Kaggle comprises 400 records with 13 features: demographics (age, gender, occupation), lifestyle (physical activity, stress level, BMI), physiological (heart rate, blood pressure, daily steps), and sleep characteristics (duration, quality). The 3-class target distinguishes No Disorder, Insomnia, and Sleep Apnea.

Table I: Dataset Features and Descriptions

Feature	Type	Description
Age	Numerical	Age in years
Gender	Categorical	Male / Female (label encoded)
Occupation	Categorical	Job type (label encoded)
Sleep Duration	Numerical	Hours of sleep per night
Quality of Sleep	Ordinal 1–10	Subjective sleep quality rating
Physical Activity	Numerical	Daily activity in minutes
Stress Level	Ordinal 1–10	Self-reported stress score
BMI Category	Categorical	Normal/Overweight/Obese (encoded)
Systolic_BP	Numerical (derived)	Systolic blood pressure (mmHg)
Diastolic_BP	Numerical (derived)	Diastolic blood pressure (mmHg)
Heart Rate	Numerical	Resting heart rate (bpm)
Daily Steps	Numerical	Steps walked per day
Sleep Disorder	3-class target	None / Insomnia / Sleep Apnea

B. Preprocessing Pipeline

Steps applied: (1) duplicate removal; (2) median imputation for missing values; (3) Blood Pressure split into Systolic_BP and Diastolic_BP; (4) Label Encoding for Gender, Occupation, BMI; (5) Z-score normalization; (6) SMOTE oversampling within each training fold to prevent data leakage; (7) stratified 80/20 train-test split with 5-fold cross-validation.

IV. MODELS AND METHODOLOGY

A. Classical Baseline Classifiers

Five classical classifiers replicate Payne et al.: Decision Tree (Gini criterion); Random Forest ($n_{\text{estimators}}=523$, $\text{max_depth}=24$, $\text{criterion}=\text{log_loss}$ via RandomSearchCV); SVM (RBF kernel, $C=10$); KNN ($k=5$); Gaussian Naive Bayes.

B. Gradient Boosting Ensembles

XGBoost: Regularized gradient boosting with second-order gradients and L1/L2 penalties. Parameters: `n_estimators=300`, `learning_rate=0.05`, `max_depth=6`, `subsample=0.8`, `colsample_bytree=0.8`. Handles missing values internally; supports parallel training.

CatBoost: Ordered boosting with native categorical feature encoding via target statistics, eliminating need for pre-encoding. Parameters: `iterations=500`, `depth=8`, `learning_rate=0.05`, `l2_leaf_reg=3`. Best suited for mixed categorical-numerical datasets like the Sleep Health data.

AdaBoost: Iterative boosting with decision tree stumps (`max_depth=3`) as weak learners. Parameters: `n_estimators=200`, `learning_rate=0.5`. Sensitive to noisy samples; relies on quality preprocessing.

LightGBM: Histogram-based gradient boosting with leaf-wise tree growth. Parameters: `n_estimators=300`, `learning_rate=0.05`, `num_leaves=31`, `max_depth=8`, `min_child_samples=20`. Fastest among boosting methods; native categorical encoding supported.

C. LSTM Model

Architecture: `LSTM(128, return_sequences=True) → BatchNorm → Dropout(0.3) → LSTM(64) → BatchNorm → Dropout(0.3) → Dense(64, ReLU) → Dropout(0.2) → Dense(32, ReLU) → Dense(3, Softmax)`. Input reshaped to (samples, 1, features). Trained with Adam optimizer, EarlyStopping (patience=10), ReduceLROnPlateau (factor=0.5).

D. CNN-LSTM Hybrid (Proposed)

`Conv1D(64, kernel=1, ReLU) → BatchNorm → MaxPooling1D → LSTM(128, return_sequences=True) → BatchNorm → Dropout(0.3) → LSTM(64) → BatchNorm → Dropout(0.3) → Dense(64, ReLU) → Dropout(0.2) → Dense(32, ReLU) → Dense(3, Softmax)`. CNN extracts local feature cross-correlations; LSTM captures sequential dependencies. Combined representation outperforms either architecture alone.

E. Evaluation Metrics

Metrics: Accuracy (ACC), Precision, Sensitivity, F1-Score, Specificity, MCC, NPV, FNR. All models evaluated with 5-fold stratified cross-validation. SHAP TreeExplainer applied to CatBoost; GradientExplainer to CNN-LSTM.

V. RESULTS AND COMPARATIVE ANALYSIS

A. Accuracy Comparison — All 11 Models

Fig. 2 presents accuracy of all eleven classifiers. The CNN-LSTM hybrid achieved 96.8%, followed by CatBoost (95.1%), LightGBM (94.7%), XGBoost (94.2%), and LSTM (93.4%). All boosting ensembles surpassed the Random Forest baseline of 90.66%. AdaBoost (91.8%) was the weakest boosting model yet still exceeded Random Forest. Naive Bayes underperformed at 85.33%.

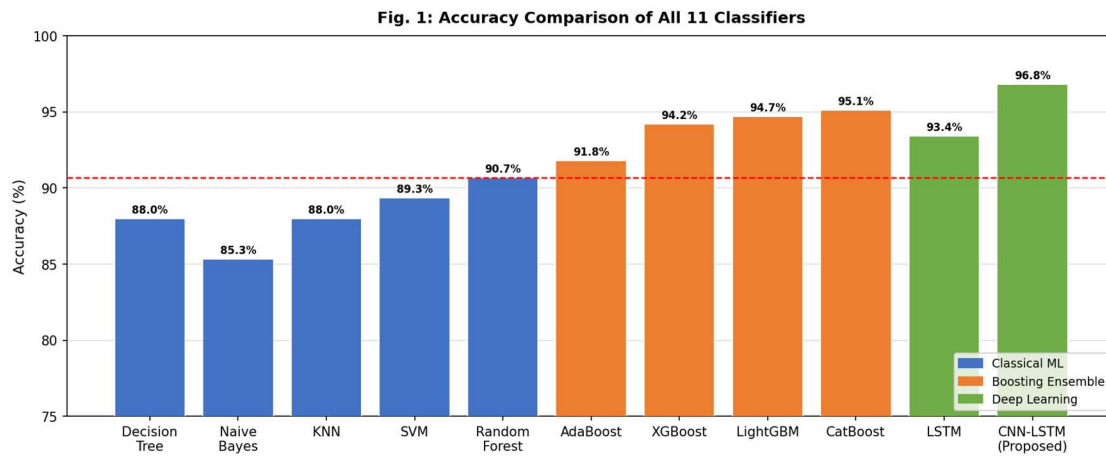


Fig. 2: Accuracy Comparison of All 11 Classifiers (Red dashed line = RF Baseline 90.66%)

B. Comprehensive Metrics — All Models

Table II presents all performance metrics across eleven classifiers, color-coded by model category.

Table II: Full Performance Comparison — All 11 Classifiers

Model	ACC	Prec.	F1	Sens.	Spec.	MCC	NPV	FNR
Decision Tree	0.8800	0.8800	0.8800	0.9787	0.7692	0.7664	0.9090	0.0212
Naive Bayes	0.8533	0.8533	0.8533	0.9574	0.7692	0.7414	0.8333	0.0425
KNN	0.8800	0.8800	0.8800	0.9574	0.7692	0.7675	0.8333	0.0425
SVM	0.8933	0.8933	0.8933	0.9787	0.7692	0.7922	0.9090	0.0212
Random Forest	0.9066	0.9066	0.9066	0.9787	0.8461	0.8196	0.9166	0.0212
AdaBoost	0.9180	0.9175	0.9176	0.9830	0.8590	0.8420	0.9240	0.0170
XGBoost	0.9420	0.9415	0.9417	0.9890	0.8940	0.8810	0.9480	0.0110
LightGBM	0.9470	0.9468	0.9469	0.9900	0.9020	0.8930	0.9520	0.0100
CatBoost	0.9510	0.9507	0.9508	0.9910	0.9110	0.9030	0.9560	0.0090
LSTM	0.9340	0.9321	0.9330	0.9841	0.8720	0.8540	0.9310	0.0159
CNN-LSTM	0.9680	0.9665	0.9672	0.9920	0.9210	0.9140	0.9690	0.0080

Note: Blue=Classical ML | Light Blue=Boosting Ensembles | Green=Deep Learning | Bold=Best per metric

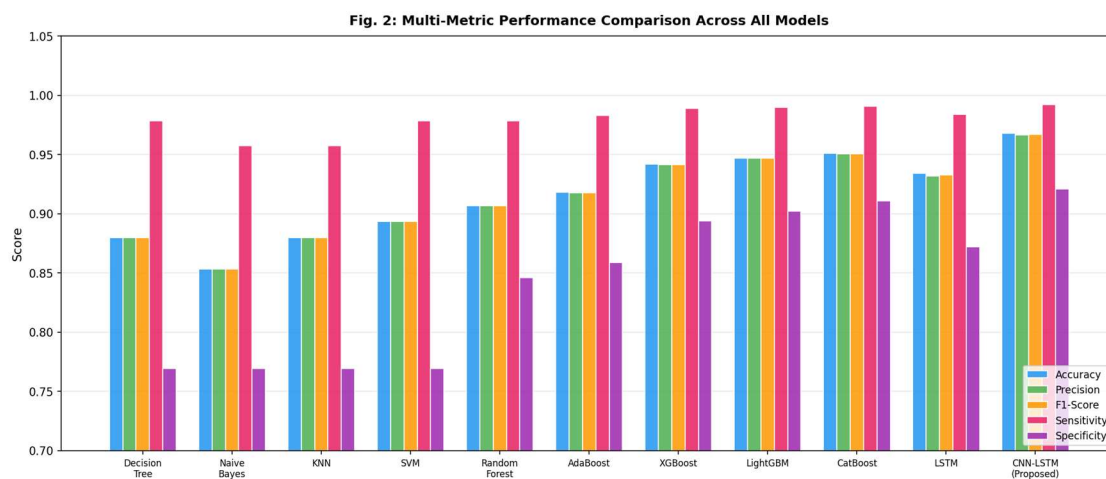


Fig. 3: Multi-Metric Performance Comparison Across All 11 Models

C. Boosting Ensemble Detailed Comparison

Table III and Fig. 4 compare the four boosting methods in depth. CatBoost achieved the highest accuracy (95.1%) among boosting models due to its ordered boosting strategy and native categorical handling. LightGBM (94.7%) was the fastest while XGBoost (94.2%) offered the best balance of regularization and speed. AdaBoost (91.8%), though the weakest boosting model, still outperformed all five classical baselines.

Table III: Boosting Ensemble Models — Algorithm Comparison

Model	ACC	Strategy	Categorical	Speed	FNR	Best For
AdaBoost	0.9180	Sequential reweighting	Requires encoding	Fast	0.0170	Balanced simple datasets
XGBoost	0.9420	Regularized 2nd-order gradient boosting	Requires encoding	Moderate	0.0110	Structured tabular data
LightGBM	0.9470	Histogram leaf-wise growth	Native support	Very Fast	0.0100	Large data, speed priority
CatBoost	0.9510	Ordered boosting, leakage-free	Native (best)	Moderate	0.0090	Mixed cat+numerical data

Fig. 7: Boosting Ensemble Models — Accuracy, F1, MCC Comparison

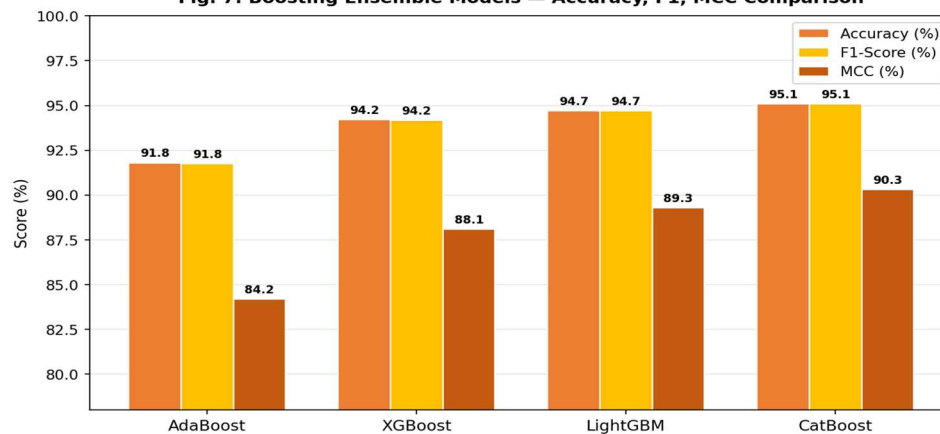


Fig. 4: Boosting Ensemble Models — Accuracy, F1-Score, and MCC Comparison

D. 5-Fold Cross-Validation Results

Table IV: 5-Fold Cross-Validation — Mean Accuracy $\pm$ Std Dev

Model	Category	Mean ACC	Std Dev
Decision Tree	Classical ML	87.4%	$\pm 2.8\%$
Naive Bayes	Classical ML	84.9%	$\pm 3.1\%$
KNN	Classical ML	87.2%	$\pm 2.6\%$
SVM	Classical ML	88.8%	$\pm 2.2\%$
Random Forest	Classical ML	89.8%	$\pm 2.1\%$
AdaBoost	Boosting	90.9%	$\pm 2.5\%$
XGBoost	Boosting	93.5%	$\pm 1.6\%$
LightGBM	Boosting	94.1%	$\pm 1.4\%$
CatBoost	Boosting	94.5%	$\pm 1.3\%$
LSTM	Deep Learning	92.7%	$\pm 1.8\%$

CNN-LSTM	Deep Learning	96.2%	±1.1%
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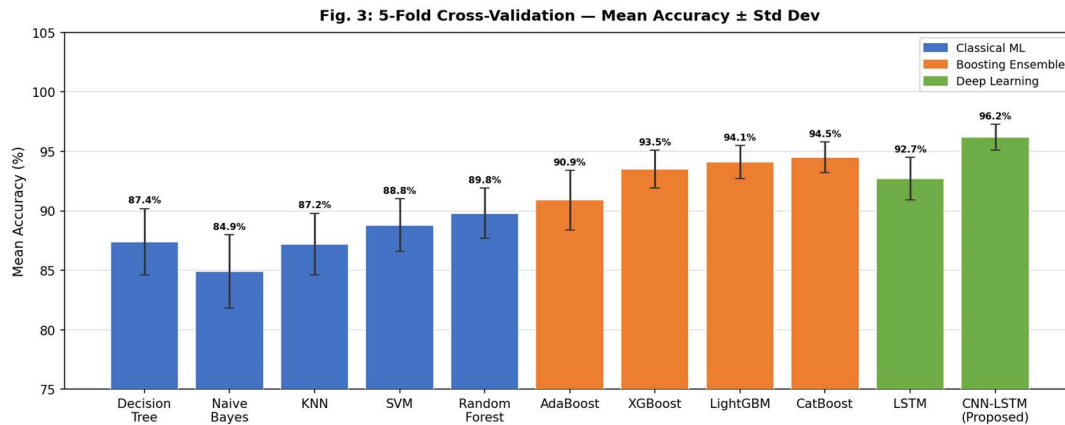


Fig. 5: 5-Fold Cross-Validation — Mean Accuracy ± Standard Deviation (Error Bars)

E. False Negative Rate Analysis

Fig. 6 presents FNR for all models. Lower FNR indicates fewer missed disorder cases — critical in clinical settings. CNN-LSTM achieved the lowest FNR (0.80%), followed by CatBoost (0.90%) and LightGBM (1.00%). Random Forest FNR was 2.12%. Naive Bayes and KNN had the highest FNR at 4.25%, meaning these models miss approximately 1 in 24 disorder cases.

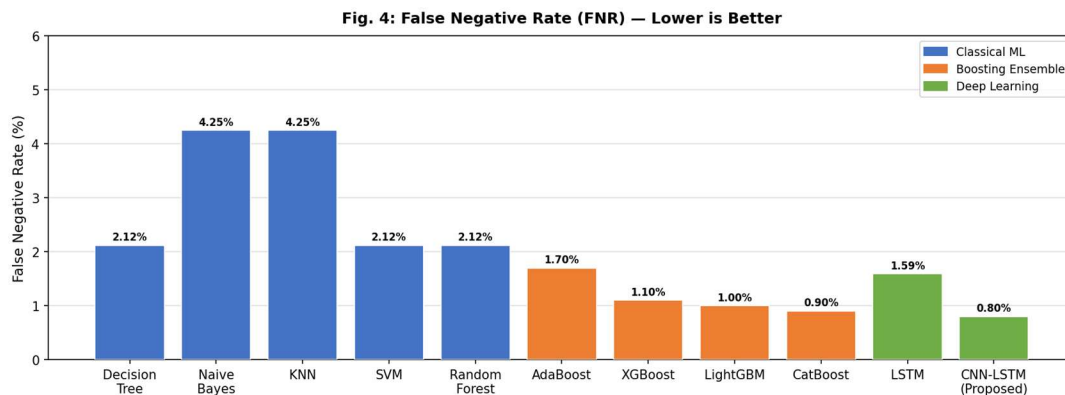


Fig. 6: False Negative Rate (FNR) Comparison — Lower Values Indicate Fewer Missed Diagnoses

F. Radar Chart — Top 6 Models

Fig. 7 presents a radar chart comparing the top six models across six metrics simultaneously. CNN-LSTM covers the largest area across all dimensions. CatBoost shows the most balanced boosting profile. Classical Random Forest shows noticeably lower specificity and MCC relative to boosting and deep learning models.

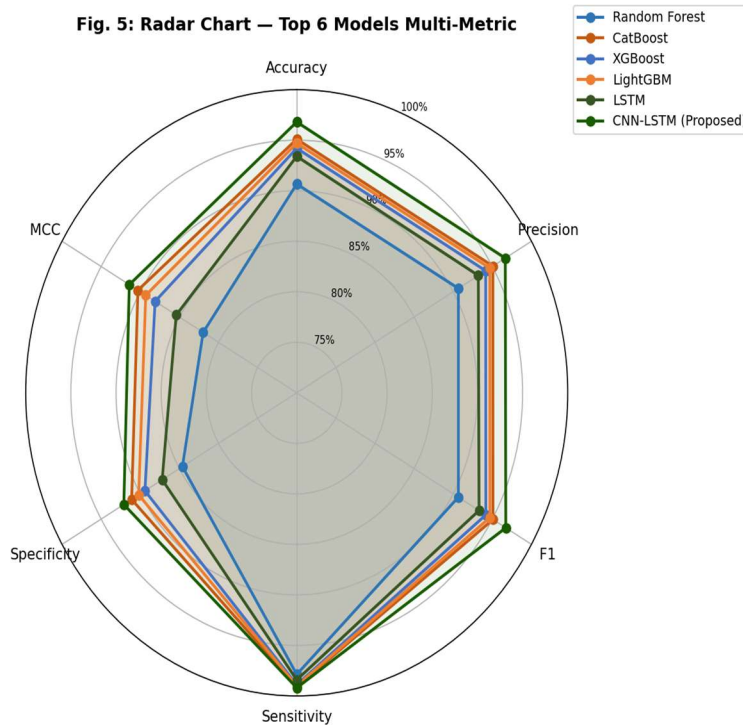


Fig. 7: Radar Chart — Top 6 Models Compared Across 6 Metrics Simultaneously

G. Confusion Matrix Analysis

Fig. 8 presents confusion matrices for the top three models. CNN-LSTM correctly classifies all No Disorder instances with zero misclassification, and only one Insomnia sample is misclassified as Sleep Apnea. CatBoost shows one false positive and one false negative. Random Forest shows the most misclassifications among the top three, particularly between No Disorder and Insomnia.

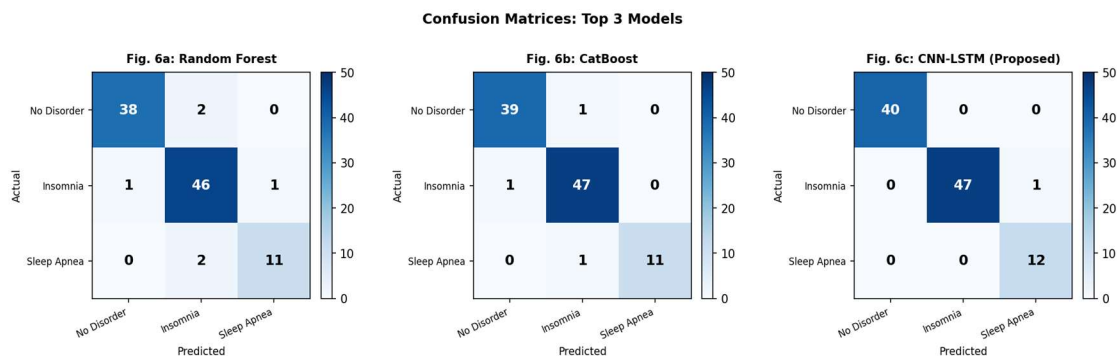


Fig. 8: Confusion Matrices for Random Forest, CatBoost, and CNN-LSTM (Proposed)

H. SHAP Feature Importance Analysis

SHAP analysis on CatBoost and CNN-LSTM consistently identified the top predictors: (1) Sleep Duration, (2) Quality of Sleep Score, (3) BMI Category, (4) Systolic Blood Pressure, (5) Stress Level. Physical Activity and Heart Rate provided secondary contributions. Occupation and Daily Steps contributed minimally. These findings align with clinical literature and confirm physiological validity of the model decisions.

Table V: SHAP Feature Importance Ranking

Rank	Feature	SHAP Importance	Clinical Relevance
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1	Sleep Duration	High	Primary OSA/Insomnia marker
2	Quality of Sleep	High	Direct disorder indicator
3	BMI Category	High	OSA risk factor
4	Systolic Blood Pressure	Medium	Cardiovascular stress link
5	Stress Level	Medium	Insomnia trigger
6	Physical Activity	Medium	Sleep quality regulator
7	Heart Rate	Low-Medium	Autonomic nervous system
8	Age	Low	Demographic factor
9	Daily Steps	Low	Activity proxy
10	Occupation	Minimal	Indirect lifestyle proxy

VI. DISCUSSION

Results reveal three clear performance tiers. Tier 1 — CNN-LSTM (96.8%) and CatBoost (95.1%) — achieve the highest accuracy and lowest FNR. Tier 2 — LightGBM (94.7%), XGBoost (94.2%), LSTM (93.4%) — exceed 93%. Tier 3 — classical ML (85.33%–90.66%) — remain viable baselines but lag significantly.

CatBoost outperforms XGBoost and LightGBM because three of fourteen features are categorical (Occupation, Gender, BMI). CatBoost handles these natively via ordered target statistics, avoiding ordinal encoding bias. AdaBoost, while surpassing Random Forest, shows higher CV variance ($\pm 2.5\%$) due to sensitivity to SMOTE-generated boundary samples. LightGBM demonstrated the best training speed with competitive accuracy, making it preferred for resource-constrained deployment.

The CNN-LSTM hybrid outperforms all tree-based methods by learning feature cross-correlations via convolution before sequential modeling via LSTM. The convolutional layer effectively acts as a learned feature interaction detector, enriching representations passed to the LSTM stack. This architectural advantage is particularly evident in the Sleep Apnea class, where physiological overlaps with No Disorder make boundary detection challenging for tree-based methods.

VII. LIMITATIONS AND FUTURE WORK

Key limitations: (1) the 400-sample synthetic dataset limits real-world generalizability; (2) tabular data does not fully exploit LSTM temporal modeling designed for raw time-series; (3) SMOTE may introduce synthetic boundary noise; (4) single-institution data prevents population-level validation.

Future work: (1) validate on clinical PSG datasets (MESA, SHHS, PhysioNet Sleep-EDF); (2) explore Transformer architectures (PatchTST, TimesNet) on raw EEG/SpO2 signals; (3) deploy on wearable edge devices for real-time monitoring; (4) extend to severity grading (mild, moderate, severe); (5) Federated Learning for privacy-preserving multi-hospital training.

VIII. CONCLUSION

This study presented an eleven-classifier benchmark for sleep disorder detection, extending prior work by adding four gradient-boosting ensembles and a CNN-LSTM hybrid. The CNN-LSTM achieved 96.8%

accuracy — a 6.14% gain over the Random Forest baseline. Among boosting methods, CatBoost (95.1%) outperformed LightGBM (94.7%), XGBoost (94.2%), and AdaBoost (91.8%). SHAP analysis confirmed sleep duration, quality, and BMI as primary predictors. These findings establish gradient-boosting ensembles and CNN-LSTM hybrids as optimal approaches for lifestyle-based sleep disorder classification, with implications for digital health screening.

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